

4(a)(i)	circles drawn around any two particles with seven (uncircled) particles in between	A1
4(a)(ii)	curve has an initial negative displacement and initial amplitude same as original curve	B1
	curve has same amplitude as original curve throughout	B1
	curve has same wavelength as original curve throughout, with constant (non-zero) phase difference	B1
4(a)(iii)	(to the) right / rightwards	A1
4(b)	$\lambda = 2 \times 0.19$ $= 0.38 \text{ m}$	C1
	$v = f\lambda$	C1
	$v = (16 \times 10^3) \times 0.38$ $v = 6100 \text{ m s}^{-1}$	A1
4(c)	$I \propto A^2$	C1
	$I = (1/2)^2 I_0$ intensity = $0.25 I_0$	A1
4(d)(i)	same frequency / wavelength / period	B1
4(d)(ii)	- a stationary wave has nodes/antinodes (and a progressive wave does not) - a stationary wave does not transfer/propagate energy (and a progressive wave does transfer/propagate energy) - different points on a stationary wave have different amplitudes (and all points on a progressive wave have the same/constant amplitude) - stationary wave has adjacent particles that are in phase (and adjacent particles on progressive wave are out of phase) <i>Any two points, 1 mark each. Allow the reverse statement for each marking point.</i>	B2